

Continual Learning on Split-MNIST

Continual Learning on Split-MNIST: Methodological Comparison and Empirical Analysis

1. Introduction

Catastrophic forgetting remains a critical challenge in continual learning (CL), where models tend to forget previously learned tasks when trained sequentially on new tasks. To benchmark and understand the practical performance of various continual learning methods, we evaluate and compare eight approaches on the Split-MNIST dataset, a widely adopted CL benchmark with five binary classification tasks.

2. Dataset and Experimental Setting

Split-MNIST Tasks:

- Task 1: Digits 0 vs. 1
- Task 2: Digits 2 vs. 3
- Task 3: Digits 4 vs. 5
- Task 4: Digits 6 vs. 7
- Task 5: Digits 8 vs. 9

Each method was trained sequentially on these tasks with accuracy measured after each task to evaluate continual performance. We report three standard metrics: Average Forgetting, BWT (Backward Transfer), and FWT (Forward Transfer).

3. Methods Evaluated

Vanilla (Baseline): No protection mechanism. Severe forgetting (~47%).

EWC: Regularization with Fisher matrix. Forgetting ~47.15%.

MAS: Output gradient-based regularization. Forgetting ~46.11%.

Replay: Sample replay, 500 samples. Forgetting ~47.15%.

A-GEM: Gradient projection with buffer. Forgetting ~47.12%.

GEM: Optimization with QP, strongest. Forgetting ~7.70%.

SI: Importance accumulation, lightweight. Forgetting ~29.25%.

4. Quantitative Summary

(Refer to table.)

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5. Visual Comparison

- Vanilla/EWC/MAS/Replay: severe early forgetting.
- GEM: flattest accuracy curves.
- SI: moderate forgetting.

6. Discussion and Recommendations

- GEM achieves the best performance but is computationally expensive.
- SI offers a strong lightweight alternative.
- Regularization and simple replay strategies have limited effectiveness on simple datasets.

7. Conclusion

GEM dominates in performance but methods like SI are practical in limited-resource settings. Future work should include more challenging benchmarks like Split CIFAR-100.

4. Quantitative Summary

Method	Type	Avg. Forgetting (%)	BWT (%)	FWT (%)
Vanilla	Baseline	47.08	-47.08	NaN
EWC	Regularization	47.15	-47.15	NaN
MAS	Regularization	46.11	-46.11	NaN
Replay	Replay	47.15	-47.15	NaN
A-GEM	Replay	47.12	-47.12	NaN
GEM	Optimization	7.70	-7.70	NaN
SI	Optimization	29.25	-29.25	NaN

Note: All methods yield NaN for FWT as Split-MNIST tasks are disjoint, hence no forward transfer is measurable.